12/03

Structure Packing

how to avoid structure padding?
1. using #pragma pack
2. using attribute pack
3. proper alignament of members

#pragma pack only works withg power of 2 values.

```
#pragma pack(2)

struct St
{
        char x;
        int y;
};
```

as pack value is 2 it allocate memory in multiples of 2

2 + 4 } 6byte

```
#pragma pack(2)

struct St
{
        char a[3];
        short int b;
        int c;
};
```

} 10byte

```
#pragma pack(4)
struct St
{
        char ch;
        int x;
};

#pragma pack(1)
struct St
{
char ch;
int x;
char y;
};
```

} 6byte

nested structure

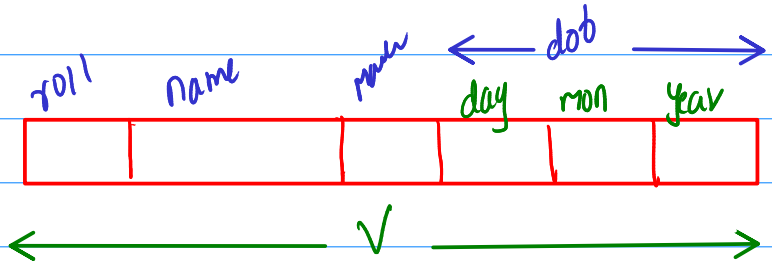
declaration structure inside a structure is called nested structure.

nested structure can be implemeted in two methods.

1. embedding structure declaration inside a structure
2. declartion variable of a structure inside a structure.

embedding:

```
struct Student
{
        int roll;
        char name[20];
        float marks;
        struct DOB
        {
                int day;
                int mon;
                int year;
        }dob;
}V;
```



V.roll
V.name
V.marks
V.dob.day
V.dob.mon
V.dob.year

selfreferential structure

Astructure whose member is a pointer to same structure is called selfreferential structure.

```
struct St
{
        int roll;
        char *name;
};
```

this is not self referential structure because name is a pointer to char

```
struct St
{
        int roll;
        char name[20];
        struct St *ptr;
};
```

this is a self referential structure

→ self referential pointer

Applications of selfreferential structures comes in link list.

Can we have same structure variable as member to structure?

```
struct ABC
{
        int data;
        struct ABC v;// error because of incomplete type
// it is incomplete because its declaration not yet completed.
};
```

Then How Pointer is acceptable?
pointer is acceptable because pointer size is 8bytes irrespective of their datatype